What Is Your Theory?

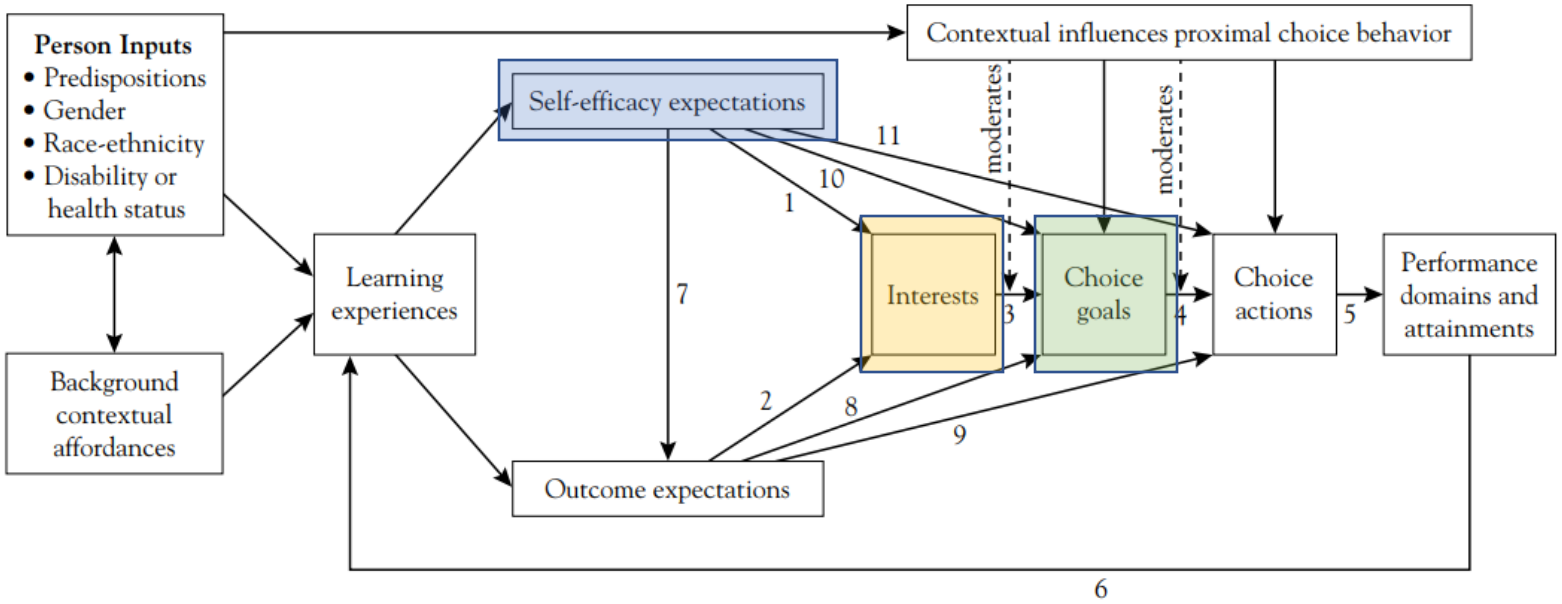
A theory is a supposition or a system of ideas intended to **explain** something, especially one based on general principles independent of the thing to be explained.

Theory -> Model

A model is a representation that specifies the causal relations between variables of interest as hypothesized by a theory.

Social Cognitive Career Theory

FIGURE 7.2. Model of Person, Contextual, and Experiential Factors Affecting Career-Related Choice Behavior



Note: Direct relations between variables are indicated with solid lines; moderator effects (where a given variable strengthens or weakens the relations between two other variables) are shown with dashed lines.

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Without a theory, statistical analysis is meaningless.

Do you have a theory that can help you understand this world or your field?

Preparation

- Install R
  - https://cran.r-project.org/bin/windows/base/
  - https://cran.r-project.org/bin/macosx/
- Install RStudio (integrated development environment; IDE)
  - https://posit.co/download/rstudio-desktop/
- Create a project for this course
- Download welcome dataset and save it in your project folder
- Run “welcome” script

Why R?

- R is the most comprehensive statistical analysis package, as new technology and ideas often appear first in R.
- R is an **open-source** that's why you can run R anywhere any time and even sell it under conditions of the license.
- It is cross-platform which runs on many operating systems. It's best for GNU/Linux and Microsoft Windows.
- In R, everyone is welcomed to provide bug fixes, code enhancements, and new packages.
- R promotes open science.

Let’s Get Started!